Pranta Bhowmik

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Academic Profile

MSc student in Georesources Management at RWTH Aachen University with research experience in GIS, remote sensing, hydrogeology, groundwater potential assessment, flood-risk mapping, and environmental quality assessment. Skilled in geospatial analysis, AHP-based spatial modelling, and machine learning, with research interests including water quality, environmental contamination, groundwater assessment, and environmental monitoring.

Education

- **MSc in Georesources Management –** RWTH Aachen University, Aachen, NRW, Germany (Apr 2026–Present)
- **BSc in Geography and Environment –** Shahjalal University of Science and Technology, Sylhet-3114, Bangladesh (2019–2024) | CGPA: 3.65/4.00

Publications

- Bhowmik, P., Laskor, M. A. H., Bhowmik, A., & Islam, M. M. (2026). Comparative assessment of flood risk using AHP and machine learning models in Kurigram District, Bangladesh. *Journal of Geospatial Science and Analytics, 2*(2), 159–186. https://doi.org/10.58524/jgsa.v2i2.139
- Bhowmik, P., Laskor, M. A. H., & Islam, M. M. (2026). Geospatial analysis of urbanization and land use/land cover change in Kishoreganj Sadar Upazila, Bangladesh (2014–2024). *Applied Geomatics, 18*, 159. https://doi.org/10.1007/s12518-026-00811-1

Research Experience

Flood Risk Mapping | May 2025 – Feb 2026

Comparative assessment of flood risk using AHP and machine learning models in Kurigram District, Bangladesh

- Developed a GIS-based flood risk map for Kurigram District using AHP, machine learning, and spatial analysis to assess and compare flood risk zones.

Groundwater Potential Assessment | Apr 2025 – Nov 2025

Groundwater Potential Assessment in the Intensive Agricultural Zones of Kishoreganj District, Bangladesh Using GIS and AHP

- Applied GIS and AHP-based spatial modelling to assess groundwater potential using hydrogeological and environmental factors.
- Manuscript under peer review at *Discover Geoscience*, Springer Nature

Urbanization and LULC Change | Apr 2024 – Jan 2025

Geospatial Analysis of Urbanization and Land Use/Land Cover Change in Kishoreganj Sadar Upazila, Bangladesh (2014–2024*)*

- Investigated urbanization and land-use/land-cover change using GIS and Remote Sensing techniques.

Environmental Quality Assessment | July 2023 – Feb 2024

An Assessment of the Environmental Quality Index in Association with Air Pollution at Sylhet Commercial Zone

- Conducted GIS-based environmental quality assessment integrating air-quality observations and spatial analysis.
- Undergraduate Research Project; manuscript prepared for journal submission.

Work Experience

Lecturer in Geography – Kishoreganj City College, Bangladesh | May 2024 – Dec 2025

- Delivered courses in Physical and Human Geography, emphasizing environmental sustainability, GIS, spatial analysis, and real-world problem-solving skills.

Research Assistant – University Research Center, SUST | July 2020 – June 2023

- Climate Migration and Land Use Transformation of the Southeastern Highland of Bangladesh — July 2022–June 2023
- Assessment of Ecological Threat to the Coastal Wetlands of Bangladesh (Sonadia Islands) — July 2020–June 2021
- Contributed to spatial analysis, Remote Sensing, survey-data integration, ecological vulnerability assessment, and mapping.

Conference Presentations

Groundwater Modelling and More 2026, Princeton University, USA | Jun 2026

- Groundwater Potential Assessment in the Intensive Agricultural Zones of Kishoreganj District, Bangladesh Using GIS and AHP — Poster Presentation

International Conference on the Role of Science and Technology towards 4IR, University of Rajshahi, Bangladesh | Oct 2023

- Public Health Appearance through GIS Techniques: A Case Study of Ward – 03, Sylhet City Corporation
- Assessment of LULC Change Using GIS Techniques: A Case Study of Sylhet City Corporation, Bangladesh
- Assessing the Quality of Life Among Individuals of Ward – 03, Sylhet City Corporation — Poster Presentation

Technical Skills

- **GIS & Remote Sensing:** ArcGIS Pro, ArcGIS 10.8
- **Programming & Data Analysis:** Python, R, Machine Learning, SPSS
- **Visualization and Design:** Adobe Illustrator, Adobe Photoshop
- **Office Tools:** MS Office

Training

- *Training on Application of ArcGIS Pro for Regional Planning* – GEE Society, SUST | 2024
- *Fecal Sludge Management in Cities* – Department of Civil and Environmental Engineering, SUST | 2022
- *Coastal and Marine Disaster: Bangladesh Perspective* – DRTMC, University of Dhaka | 2022

Leadership and Service

- **Sports Secretary and Publicity Secretary**, GEE Society SUST | 2022–2024
- **Publication Secretary**, YouthMappers at SUST | 2022–2023
- **Organizational Staff Coordinator**, Hult Prize, SUST | 2022–2023

Languages

- English: Professional Working Proficiency — IELTS 6.5